SPE-226980-MS

KAUST Geothermal Exploration Project (K-GEP) – A Promising Solution for Green, Clean, Sustainable Energy Provision

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

In arid countries with a continuously growing population, affected by depleting fossil aquifers and stressed by the impact of climate change, the dependency on fossil fuel consumption for desalination and cooling may be reduced using medium-enthalpy geothermal energy. By harvesting geothermal resources, clean and sustainable energy for desalination, district cooling, and brine mineral extraction may be targeted. In our K-GEP project, we aim to demonstrate a comprehensively integrated approach. We started to explore the Kingdom's geothermal potential by drilling a shallow pilot hole on KAUST campus, located along the Red Sea shore. The pilot hole reached a depth of ~400m and provided an opportunity during drilling and subsequent logging to acquire a variety of data ranging from fiber optic distributed acoustic and temperature sensing (DAS & DTS), seismic while drilling, and multi-scale ambient-noise seismic monitoring. Drill cuttings reveal a complex heterogeneous lithology varying between sandstones, claystones, gypsum, anhydrite, and dolomite intervals. Some of these are challenging for drilling. A major unconformity from the Plio-/Pleistocene Lisan group to the Mid-Miocene Kial formation is interpreted at a depth of ~75m. Some of the layers described by wellsite geology can also be resolved using DAS in combination with advanced seismic ambient-noise processing methods. Repeated temperature measurements along the hole (both logging and DTS) confirm an average heat gradient of ~31°C/km (once downhole temperatures had recovered from the drilling process). Although the Red Sea Shore is only a lateral distance of ~1.1 km away from the well location, an aquifer system at the bottom of the well shows a distinct chemical signature different from sea water (including elevated lithium concentration). Our preliminary results suggest that the K-GEP project has the potential to prove the feasibility of low to medium-enthalpy geothermal development in KSA based on cascaded energy extraction.

Introduction

Like many other countries, the Kingdom of Saudi Arabia is facing the challenge of rapidly increasing energy demand for its fast-growing population and expanding industrialization. This specifically applies to the generation of drinking water and cooling. Unfortunately, seawater desalination and electricity generation for air conditioning are still based on burning fossil fuel, thereby compounding climate change (Odnoletkova and Patzek 2021). The utilization of renewable green energy is just starting, with main efforts concentrated on solar PV and, to a lesser degree, on wind energy. Tapping into geothermal energy resources in Saudi Arabia has so far been largely untested despite the Kingdom's promising high potential for its development (Vahrenkamp et al. 2020; Vahrenkamp et al. 2021; Ye al. 2023, Riera et al, 2024; Ezekiel et al, 2024; Yalcin et al, 2024).

Yet-to-be-published data and a review of geological settings indicate that heat flow is slightly above the global average over vast areas of the Arabian Peninsula by exceeding a gradient of 30 °C/km. Furthermore, aquifers are expected to exist at depth that would allow the extraction of fluids with medium enthalpy heat (Vahrenkamp et al. 2020; Vahrenkamp et al. 2021). This is especially true for both the west and east coast, but also for central areas of the Arabian Peninsula, which host a number of urban centers and industrial developments with high demand for water and cooling.

The Kingdom's Vision 2030 sets the ambitious goal to generate a substantial portion of this energy from alternative sources: solar, wind, and geothermal. However, despite its obvious advantage of being a 24/7 baseload energy source (Bolson et al., 2022), geothermal energy has yet to receive the necessary attention and support with the main reasons summarized above. A successful geothermal development will impact Saudi Arabia enormously since the Kingdom is the fastest growing electricity consumer in the Middle East (66 GW in 2016), with a projected need to increase to 120 GW by 2032 (Jadwa, 2016). Electricity generation for air-conditioning and water desalination already consumes over 2 million barrels of hydrocarbons each day in peak-demand summer months, or about 20% of Saudis' daily hydrocarbon production (Nachet & Aoun, 2015; Howarth et al., 2020).

KAUST geothermal exploration project pilot (K-GEP)

The focus for the K-GEP project is on the Kingdom's west coast, where aquifer sequences are present in deep Tertiary rift basin sedimentary layers that formed during the Red Sea opening in the early Eocene (Ye et al. 2002; Hughes and Johnsson 2005; Tubbs et al. 2014). One prime exploration site is the campus of King Abdullah University of Science and Technology (KAUST), located ~80 km north of the city of Jeddah. K-GEP's targets are medium enthalpy systems consisting of brine aquifers situated within pre- and syn-rift clastic sedimentary layers. Granitic basement depth is not exactly known but estimated to be at least 5 – 6 km. Unpublished data show a temperature gradient of ~32°C/km. Given an average annual surface temperature of 30°C, approximately 100°C is reached at 2,200m depth and 150°C at 3,750m depth. Figure 1 sketches the location and local subsurface geology of the KAUST site, indicating major basin bounding and smaller subsidiary, inter-sedimentary layer normal faults.

K-GEP's main purpose is to demonstrate direct-use geothermal-energy extraction from permeable rift-basin sediments along KSA's Red Sea shoreline and to establish convincing proof that this renewable natural resource can be scaled to become a part of the Kingdom's energy mix. Key objectives are to develop fit-for-purpose drilling techniques, describe and characterize subsurface reservoir properties, including subsurface plumbing, fluid flow management, and thermal behavior in heterogeneous rift basin sediments. In this context, Figure 1 sketches a geothermal well doublet reaching ~3,750m (presumed to be near the basement top). Without requiring reservoir stimulation, geothermal brines would be produced from one or more aquifers to surface at a sufficient flow rate for sustained heat extraction and direct use (not necessarily for generating electricity) before being pumped back into the same aquifer, where they gradually reheat and then be re-produced from the production well.

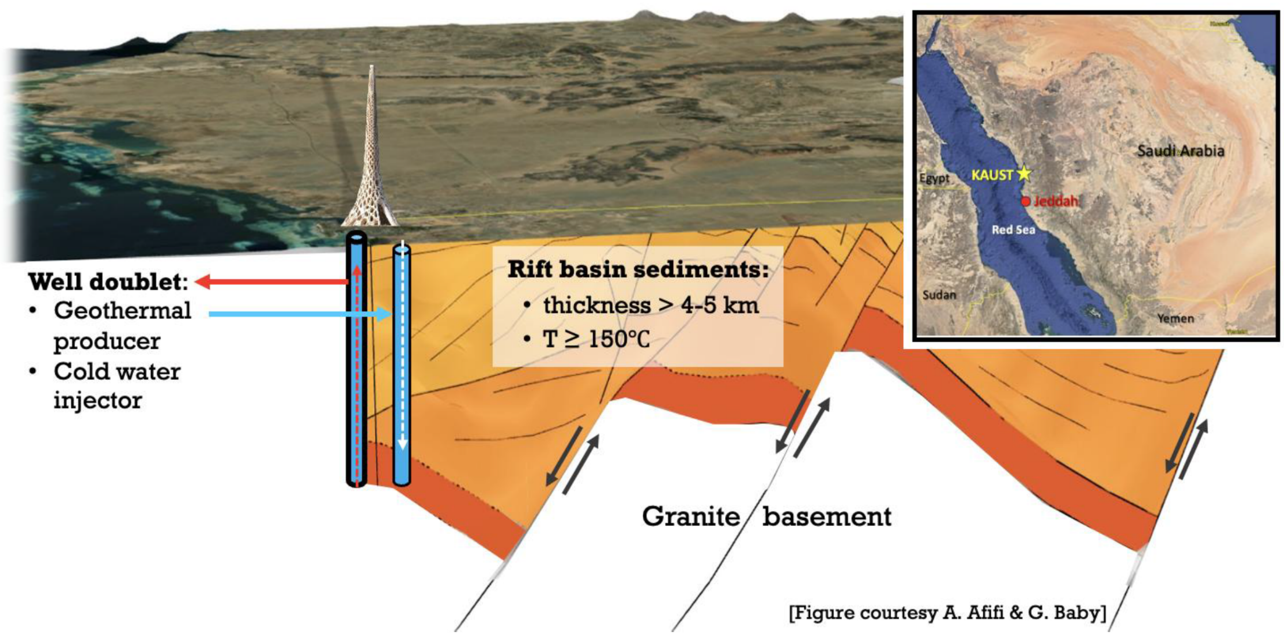

Figure 1—Graphical sketch depicting the so-called KAUST Beacon as a symbol of KAUST's location at the Red Sea shoreline (infer regional context of location from the inset). The subsurface sketch (courtesy A. Afifi & G. Baby) highlights main lithologic layers as well as major basin bounding and smaller subsidiary, inter-sedimentary layered normal faults typical for rift basins along the Red Sea coast. Symbolically we also show a producer and injector geothermal well doublet penetrating the sedimentary layers down to the top of the basement. The box indicates expected total sediment cover thickness and temperatures for exploration target depths.

K-GEP is planned in four phases:

- **Phase 1**: A 400 m pilot well for initial testing, data collection, and monitoring & surveillance purposes. Acquired data include ambient-noise seismic recordings, temperature profiles, distributed acoustic sensing (DAS), and distributed temperature sensing (DTS) from fiber optics.
- **Phase 2**: Construct the first deep well to a depth of 3,000m to 3,500m TVD with the main objective to de-risk exploration activities, acquire valuable drilling experience and downhole data, characterize saline aquifer layers with acceptable flow rates for heat extraction at the surface, and conduct fit-for purpose reservoir simulation to decide on spacing for next well.
- **Phase 3**: Construct the second deep well to a depth of 3,000m to 3,500m TVD with the objective to implement a geothermal heat extraction system utilizing integrated and innovative technologies both for sub-surface as well as surface heat exploitation.
- **Phase 4:** Upscale surface facilities and connect to KAUST campus infrastructure; implement long-term monitoring and establish a research observatory and technology innovation platform to foster and advance the Kingdom's geothermal energy developments.

K-GEP Phase 1 – drilling of a shallow pilot well for reconnaissance, surveillance, and monitoring purposes – was funded by KAUST in 2024. This well – named *K-GEP-1* – is situated approximately 1.1km inland from the Red Sea shore in an undeveloped part of the campus and thus does not interfere with academic infrastructure and residential areas. In contrast, the site is strategically located close (i.e., within less than 500m) to three key facilities that are envisaged to be potential users of geothermal energy in the future. These facilities are KAUST's desalination, chiller, and microalgae production plants. Figure 2a provides an aerial map for orientation. The well reached a total depth (TD) of 392m below ground level and is fully cased and cemented (i.e., no formation contact). Figure 2b shows the well schematic with conductor casing (13 ⅜ inch) down to 25m, an intermediate casing (10 ¼ inch) point at 195m, and a final casing string (7") down to TD. Before casing and cementing operations took place, the following open hole log data were acquired: gamma ray (GR), single arm caliper for hole diameter, resistivity, and temperature. We show the log responses also in Figure 2b and discuss them in combination with drill cuttings analysis further below. Outside of casing, we deployed and permanently cemented in place single- and multi-mode fiber optic cables for distributed acoustic and temperature sensing (DAS and DTS) as well as a capillary tube, which allows fiber optic cable installation and replacement at later times. The fiber cables, as well as the capillary tube, reached a depth of 333m and are then looped around the 7" casing without any sharp bends to avoid damage or signal losses. Thus, the bottom 59m of the hole are on purpose without a capillary tube and fiber cables. This provides the option to perforate the hole in the future and gain access to the ambient formations across this interval. After the well was cased and cemented, we opted to drill a 5m rathole and take a 10-liter formation water sample from the bottom of the hole. Following this process, we plugged the well at 380m depth and pumped cement to backfill the hole to a depth of 372.5m. Results from the water sample analysis are discussed below.

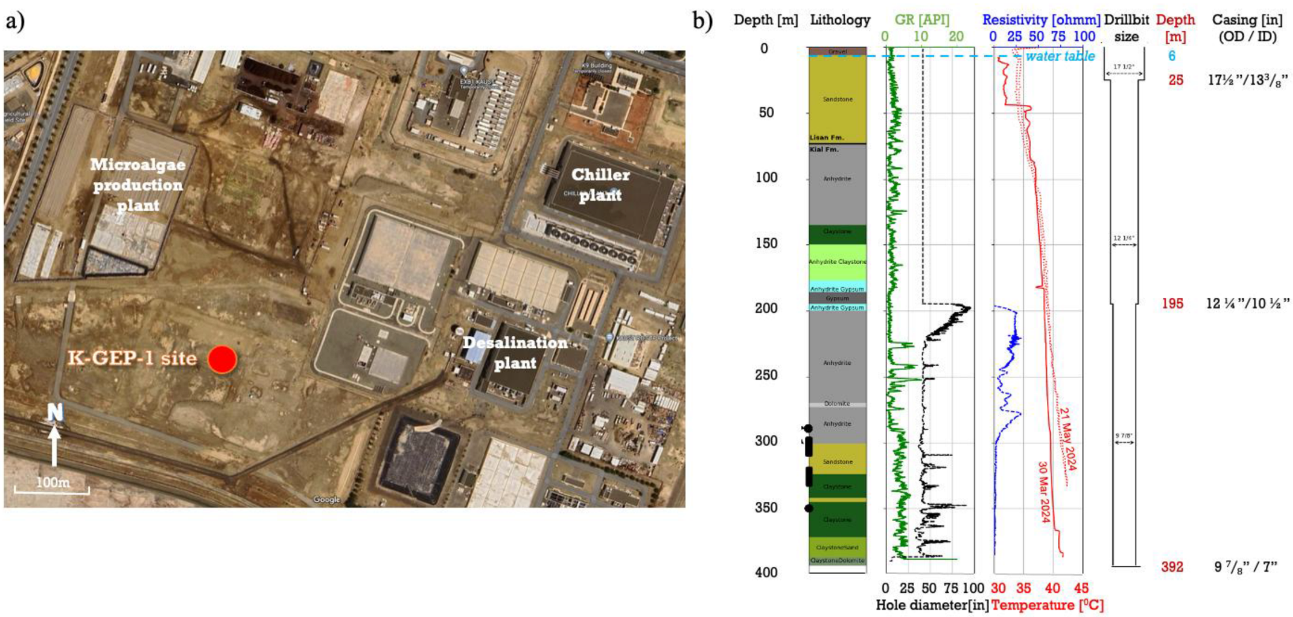

Figure 2—a) Aerial map of the K-GEP-1 well site location. It is strategically positioned within less than 500m to three potential key users of geothermal heat (as labeled). b) Lithology (left track), gamma ray (GR; green) and caliper (hole diameter; black) logs (second track from left), resistivity log (blue) and temperature profiles (red; solid line =log; dashed line = distributed temperature sensing; third track from left) as well as bit and casing depths/ sizes and depth to water table (right tracks) for K-GEP-1. The solid, black dots to the left of the lithology track indicate zones along which challenges such as tight hole or/and (partial) losses were encountered during drilling.

Drill cuttings and open hole log analysis

Drill cuttings retrieved at 3m-intervals reveal a heterogeneous lithology varying between sandstones, claystones, gypsum, anhydrite (some interlayered with chert), and dolomite units. This variable lithology is reflected by the GR and resistivity log responses, indicating various degrees of clay and pore fluid salinity contents (Figure 2b). The resistivity log only has credible data between 195m and 300m depth because it was not run shallower above the 10 ¼ inch casing shoe, and below 300m the tool malfunctioned. Based on the cuttings analysis, we interpret a major unconformity from the Plio-/Pleistocene Lisan group to the Mid-Miocene Kial formation at ~75m depth. This is at least 25m shallower than expected based on regional experience and interpretations of unpublished seismic data, but we note that the latter are of insufficient resolution for precise depth prognoses. Thus, 25m variability is within the inaccuracy level of the data.

Drilling was challenging along some intervals with the drill crew experiencing numerous tight holes, stuck pipe and (partial) loss events. The depth where some of these incidences occurred are highlighted as black dots in Figure 2b. Further, several sections have enlarged hole diameters as evidenced by the caliper log. The most challenging conditions occurred directly below the 10 ¼ inch casing shoe from 195m to ~225m depth where the hole is severely enlarged. Partial losses resulting in insufficient mud weight caused the formation consisting of anhydrite interlayered with softer chert to wash out. These problems were eventually overcome by mud conditioning mud via adding bentonite and carboxy methyle cellulose (CMC) to increase viscosity and as filtration agent. This allowed drilling to progress.

Temperature profiles from logging and DTS data

Fiber-optic DTS technology has become a transformative tool in geothermal well monitoring, offering continuous, high-resolution temperature measurements along the entire fiber length. This capability is particularly valuable in geothermal applications, where it enables real-time observation of thermal behavior, fluid movement, and reservoir dynamics (Khankishiyev et al. 2024). In the *K-GEP* pilot well, a 50/125 µm multimode optical fiber was deployed in a U-shaped configuration, with a loop and turn point at the TD of ~333 m, as previously described. This configuration allows for double-ended DTS measurements, which are beneficial for compensating for the wavelength-dependent differential attenuation of the Stokes and Anti-Stokes Raman backscattered signals (Ashry et al. 2022). Figure 3 displays representative DTS data acquired after the well had thermally stabilized following drilling operations. Figure 3a shows a 3D surface plot of 15 consecutive DTS traces, each representing a 2-minute averaging interval for the Stokes and Anti-Stokes Raman signals. These measurements were acquired over a short time span to verify the thermal stability of the wellbore environment. The minimal variation observed along the time axis confirms that the downhole temperature profile had reached a steady state, ensuring that the recorded temperature distribution reflects the true geothermal gradient rather than transient thermal effects from drilling or fluid movement. The average temperature profile derived from these 15 traces is shown in Figure 3b, providing a smoothed representation of the geothermal gradient. As shown in Figure 3b, we observe a consistent but non-linear increase in temperature with distance along the fiber-optic cable until ~333 m, which marks the maximum depth of the cable. In particular, from ground level to about 210 m, the profile changes its gradient. This reveals a heat flux system that appears to be governed by both shallow convection and deeper conduction. As we discuss below, some of the lithological layers found during drilling appear to correlate with the transition between these different heat flow regimes. Below 210 m depth, the gradient becomes more linear. Since the optical fiber is U-shaped with a turnaround point at the total depth, the temperature profile in Figure 3b increases from the surface (Depth = 0) to the maximum value at the bottom (Depth = 333 m), then symmetrically decreases along the return leg of the fiber until reaching the surface again. In other words, the temperature profile is mirrored along the x-axis at the turnaround point, Depth = 333 m. At a depth of 333 m (bottom loop of the fiber cables), the temperature reaches 42.3 °C. In Figure 3c, we display several temperature profiles measured between May 2024 and March 2025. Especially the uppermost ~10 m are affected by seasonal temperature variations. To derive a temperature gradient for our well site, we fit a linear trend from the bottom ~133 m back to the surface. We arrive at an average ground level temperature of ~32 °C. With this, we derive an average temperature gradient of 31 °C/km, which is within published global ranges of gradients for sedimentary rift basins (Kolawole and Evenick, 2023).

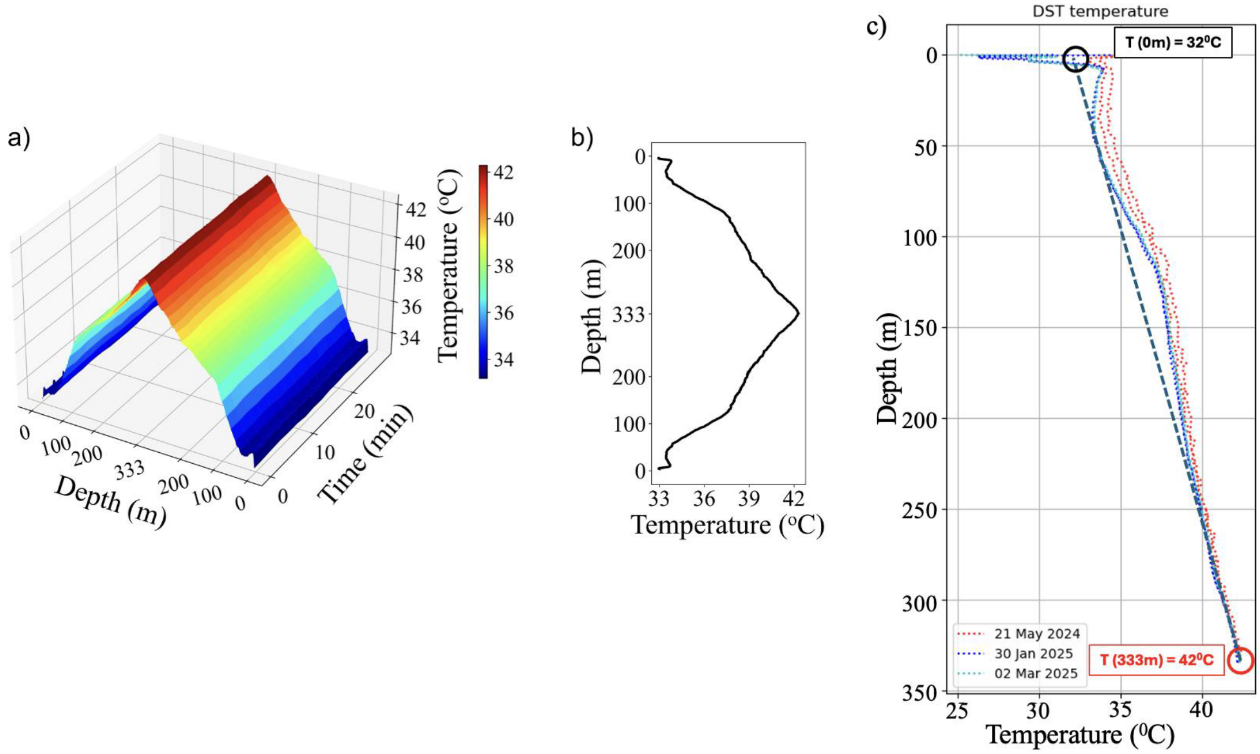

Figure 3—a) DTS temperature profiles acquired from 15 consecutive 2-minute measurements. b) Averaged profile showing a gradual increase to 42.3 °C up to a depth of ~333m along the length of the fiber-optic cable. c) Repeated temperature measurements from DTS between May 2024 and Mar 2025.

Passive multi-scale and multi-method seismic monitoring

Passive seismic monitoring is a non-intrusive, cost-effective, and environmentally friendly method to infer subsurface structures and their (potentially time-dependent) rock physical properties around geothermal wells. From inspecting the surface and very shallow (2-3 m) geology, the main challenge for seismic monitoring at the *K-GEP 1* site is the unconsolidated surface layer of sand and construction debris that strongly affect seismic-wave propagation. To overcome this challenge, we deployed various types of seismic sensors targeting different spatial and frequency-resolution scales to monitor seismic-wave energy from drilling and ambient-noise sources. The seismic monitoring system includes four 3-component broadband seismic sensors, an array of vertical-component STRYDE-nodes, and finally a permanent deep-borehole 3-component borehole seismometer (at ~270 m depth) installed after borehole completion. The broadband sensors have a bandwidth of 0.02 – 50 Hz and were deployed over a period of 5 to 18 months before drilling and are arranged in a crossed pattern at ~100 m distance from the well site. The STRYDE array was deployed in a line towards the NNE. We utilized 89 nodes spaced regularly at 2 m and a total offset of 176m. Figure 4a displays the geometrical configuration of the various systems deployed. We acquired seismic-while-drilling (SWD) data for about one month.

To extract surface waves and measure dispersion curves (target range: 5 - 15 Hz) we applied seismic interferometry to a 7-day portion of the SWD-data. The resulting dispersion curves are then used to estimate a 2D near-surface shear wave velocity model down to 20 m depth. Broadband seismic data were used to measure horizontal-to-vertical spectral ratios (HVSR). This allows us to estimate site-response transfer functions and extract shear-wave velocity profiles. We first computed the HVSR for 14 days each month, then stacked the HVSR of 5 months to obtain the final HVSR of each station. The HVSR curves show two main peak frequencies ~2 Hz and 6 Hz. By jointly inverting the HVSR and dispersion curves, we derive a 1D shear wave velocity profiles down to 150m. Figures 4b – f highlight the main data processing and analysis steps.

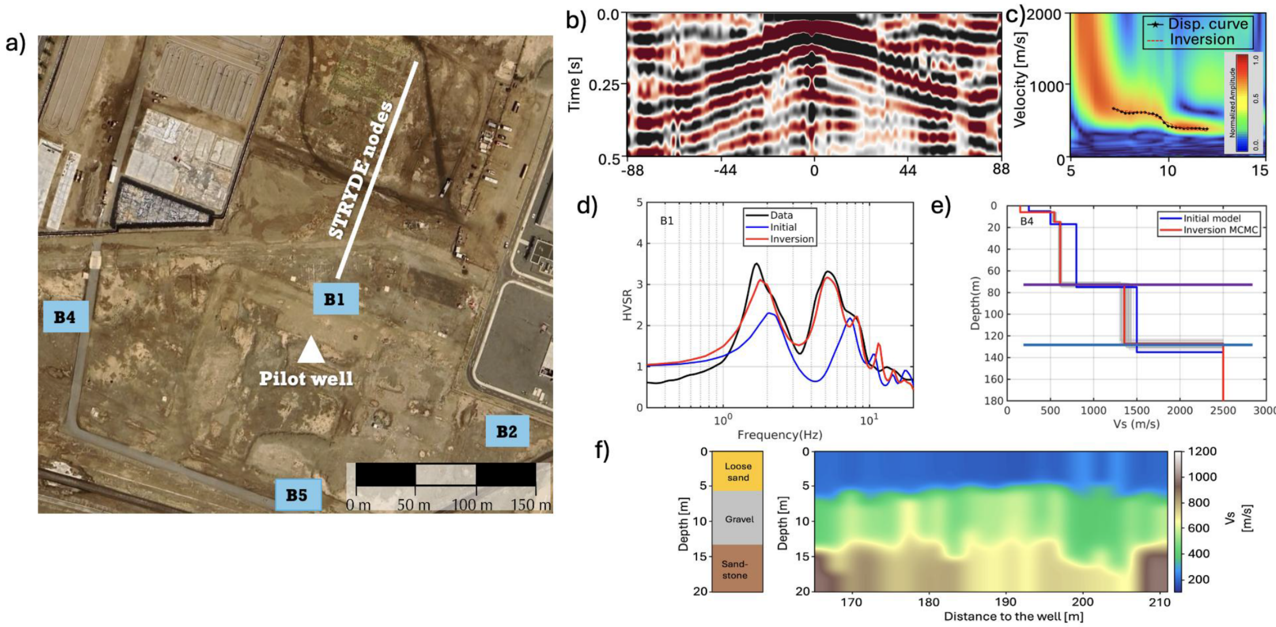

Figure 4—a) Map of the K-GEP 1 site, showing the drill location (green triangle), sites four broadband stations (blue squares) and one of the STRYDE node arrays (red line). b) Correlogram along the nodes, with the virtual source in the center; negative offsets towards the south. c) extracted dispersion curves. d) HVSR ratios at broadband site B1. e) Example for HSVR-inferred vertical shallow structure at site B4. f) simplified lithology (left column) and inverted 2D shear-wave profile from interferometry (adapted from Li et al, 2025)

Velocity profiles below broadband seismic stations and the closest STRYDE-node profile are consistent in the depth ranges of overlap. Moreover, the shear wave profiles agree with the lithology interpreted from drilling cuttings, comprising a major unconformity from the Plio-/Pleistocene Lisan group to the Mid-Miocene Kial formation at a depth of ~75m (Figure 4e). We demonstrate that a multi-scale seismic monitoring system can effectively reveal the subsurface structure of the sites of interest.

Moreover, temporary broadband stations installed for several months together with a deep-borehole seismic station allow not only to measure time-dependent subsurface rock-physical variations (e.g., due to seasonal climatic variability or fluid infiltration), but also to record nearby seismic events that may pose a hazard to the site. Such multi-scale and multi-method seismic monitoring techniques are key to holistically de-risk any geothermal site.

Water sampling and lithium potential

An initial assessment of the lithium potential of Saudi Arabia suggests that rift-basin brines should have a high potential for elevated lithium content and perhaps also other metals (Ortiz and Vahrenkamp, 2025). To test this hypothesis, water samples were collected from the bottom of the well and compared to Red Sea seawater (Table 1).

Table 1—ICP-OES results for some water samples in Saudi Arabia.

Sample	Li [ppm]	Mg [ppm]	Ca [ppm]	MG/Ca []	Sr [ppm]	Mn [ppm]	Fe [ppm]	Na [ppm]	K [ppm]
Red Sea	0.21 (±0.01)	1,487 (±6)	410.9 (±1.3)	3.62	4.78 (±0.02)	< 0.01	< 0.01	12,574 (±45)	647.8 (±2.2)
KAUST well (400m)	0.76 (±0.04)	2,376 (±0.34)	8,620 (±48)	0.27	140.0 (±0.5)	4.67 (±0.01)	1.73 (±0.03)	25,238 (±15)	359.3 (±0.8)

The Red Sea water has a higher lithium concentration than seawater from a normal marine environment; this is due to the arid conditions of the Red Sea, leading to higher evaporation and, therefore, higher concentrations of non-reactive metals.

The second sample is water from the KAUST exploratory geothermal well. This sample reveals an elevated salinity level and a lithium content of more than three times that of Red Sea seawater. Other elements such as strontium, iron, and manganese are also significantly enriched compared to seawater. In contrast, the Mg/Ca ratio is much reduced compared to that of seawater, possibly due to the incorporation of magnesium from the brine into the formation of diagenetic dolomite. Alternatively, lithium and other metals are already being leached by the groundwater from the rift basin sediments.

Discussion and summary

The nearly 400m deep pilot well was a successful endeavor for KAUST and geothermal developments in Saudi Arabia. Although no actual geothermal heat is harvested from this well, we achieved all our defined technical objectives. Fiber optic cables are installed, fully functional, and have already acquired unique data sets capturing in real time various well operations, temperature profiles, and more. Along with broadband seismic surface stations and other sensors, we are in a position to acquire, process, and analyze data that were and are continuing to be recorded and that will illuminate the Earth below our feet in KAUST. Repeated temperature measurements indicate an average gradient of 31°C per km – quite typical for most parts of the Red Sea Basin (the region around Jizan near the border with Yemen being an exception; there, geothermal temperature gradients are appreciably higher as unpublished data show).

The water sampling operation collected ~30 liters of formation water from the bottom of the well. Even though the well was drilled less than 1 km from the Red Sea shoreline, preliminary geochemical analyses show a brine composition different from seawater, including a three-fold increase in lithium concentration, elevated salinity, and enriched metal content. This indicates diagenetic processes enriching pore waters through mineral leaching and provides further impetus to our plans of drilling far deeper as a next step in our geothermal & lithium mining exploration.

We also installed a 3-component broadband seismometer at ~270m depth for continuous passive seismic monitoring of ambient ground vibrations and seismic activity in general (local micro-seismicity and regional events from the Red Sea and surrounding regions). Future planned research includes fiber optic technologies, further acoustic and temperature measurements, and more. In summary, we now have a densely instrumented pilot well on campus that serves as an underground observatory and platform for (geothermal) research and education for many years to come, allowing also for implementing and testing multi-purpose de-risking strategies.

As a next step, we plan a doublet exploration well system to test circulation and potential subsequent exploitation. This system enables critical follow-up research and facilitates connecting the K-GEP pilot to the university's facilities for desalination and chilled water production. This would make KAUST the first academic institution in Arabia to undertake such an endeavor. Our plans are not limited to exploring the subsurface for heat and aquifers strong enough to maintain long-term exploitation; instead, we plan to design and integrate surface facilities for optimum energy and mineral extraction from subsurface brines (lithium, etc.) to achieve successful economic use of geothermal energy. We envision not only substantial gains in economically maturing geothermal energy utilization but also training the next generation of geothermal engineers to support the Kingdom's Vision 2030 and associated carbon pledges.

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